BUSA8090 Assignment 2

Visual Analysis of Queensland Road Crashes: Trends, Severity, and Contributing Factors

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1. Introduction

This report analyses Queensland's road crash data using data visualization to identify temporal, spatial, and severity patterns that support road safety decisions. Visualisations were developed to explore trends over time, high-risk regions, and how severity varies by hour and crash type. The dataset, obtained from the Queensland Government Open Data Portal, was cleaned to ensure integrity, removing missing or inconsistent records. Tableau was used to create time series, heatmaps, and stacked bar charts, offering a comprehensive view of crash severity. Ethical principles such as fairness, accuracy, and transparency guided the process. The findings provide actionable insights to help policymakers design targeted, data-driven road safety interventions.

2. Data Preparation and Cleaning

Before conducting the analysis, several data preparation steps were completed to ensure the dataset's accuracy and consistency. The dataset was loaded into Jupyter Notebook for cleaning.

- Categorical fields were reviewed for misspellings or inconsistencies, but no errors were found.
- Records labelled as "Property Damage" were excluded, as this category ceased being recorded after December 2010; this step removed 87,432 rows to maintain consistency across the time series.
- The Loc_Post_Code field was standardized by converting it to string format, and one record with an unknown postcode was eliminated.
- Additionally, 376 records lacking latitude and longitude data (0.12% of the dataset) were removed to ensure accurate geographic analysis.
- Date fields were converted into proper datetime formats, enabling extraction of year, month, and hour for temporal analysis.

These cleaning processes ensured that the final dataset was accurate, complete, and suitable for generating reliable visualisations across all areas of analysis.

3. Visualisations and Analysis

3.1. Crash Patterns and Trends Across Time

To explore temporal trends, three visualisations were developed. Figure 1 presents a time series line chart showing total crashes from 2001 to 2023. A time series chart was selected to effectively capture long-term trends and fluctuations. The data shows a peak in 2007, followed by a notable decline after 2008, stabilizing around 12,000 crashes by 2015–2020. Although recent years show slight rebounds, total crashes remain below pre-2010 levels. Improvements may reflect safer vehicles, stricter enforcement, infrastructure upgrades, and public safety campaigns.

Figure 2 displays total crashes by month, using a line chart to highlight seasonal fluctuations. Crashes increase steadily from January, peaking in July–August, then decline towards year-end. These mid-year peaks may correspond to higher road use during school terms, tourism, and winter driving conditions.

Figure 3 presents a multi-series time plot, breaking down crashes by severity level. Minor injuries and medical treatment crashes show clear long-term declines, while hospitalisations reveal a slight upward trend. Fatal crashes remain relatively stable over time. This indicates that while general crash volume has improved, severe outcomes persist, requiring targeted behavioural and environmental interventions.

Overall, the time series analysis reveals both positive progress and ongoing challenges. Although total crashes have declined since 2008, stable fatality levels, seasonal mid-year peaks, and diverging severity trends highlight areas where further policy measures may be needed to address the most severe crash outcomes.

Figure 1: Time Series of Crushes

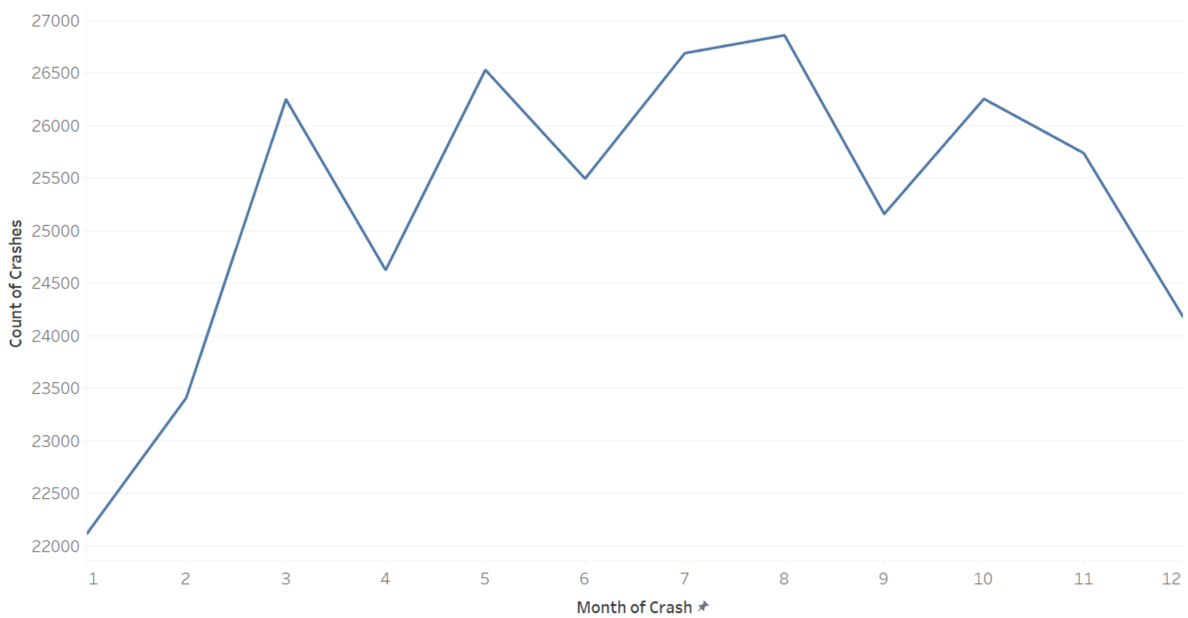

Figure 2: Total Crushes by Month

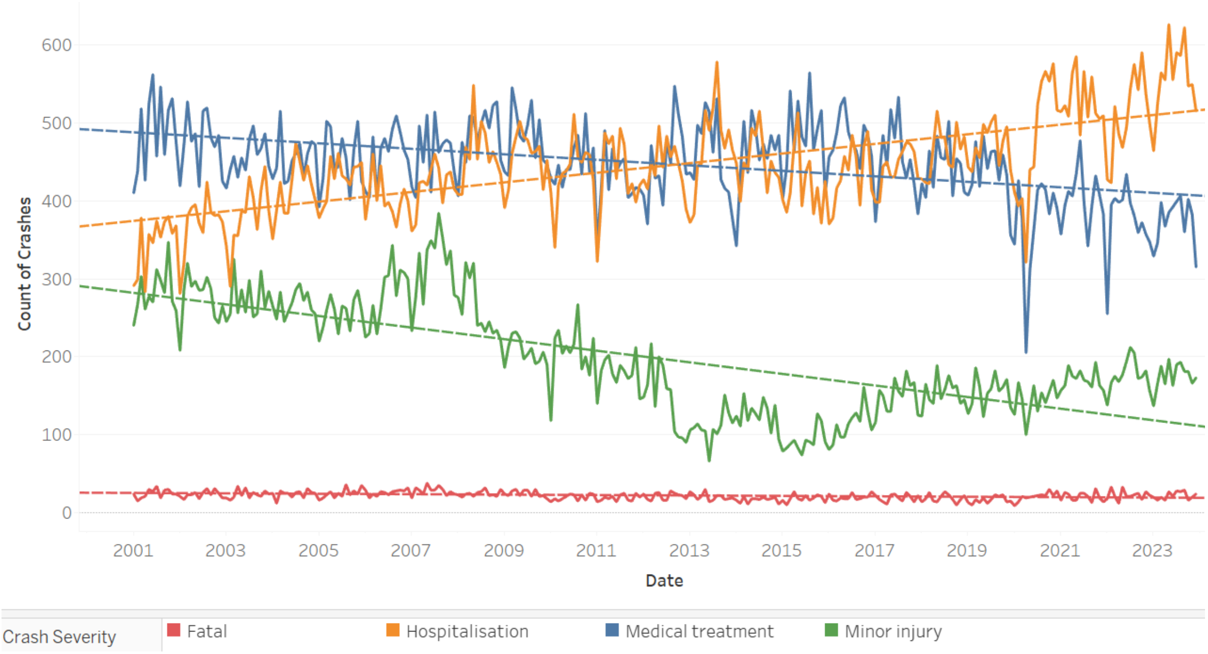

Figure 3: Crushes by Severity by Date

3.2 Fatal Crash Distribution and High-Risk Locations

To better understand the spatial distribution of fatal crashes, several visualisations were developed combining geographic, regional, and temporal perspectives.

Figure 4 presents a choropleth map showing fatal crash casualties across Queensland postcodes. The map uses a green-to-red colour scale to reflect fatality counts, helping identify high-risk zones. Spatial analysis shows that fatalities occur broadly across both urban and rural areas. South East Queensland (SEQ), including Brisbane and the Gold Coast, shows higher fatality counts, likely linked to population density and traffic volumes. However, several rural areas in Central and Northern Queensland also report significant fatalities, suggesting that factors such as high-speed zones, long travel distances, remote road conditions, and slower emergency response times may contribute to severe outcomes in these regions.

Figure 5 focuses on the top 10 postcodes with the highest fatal crash counts, providing a more targeted view of Queensland's most critical hotspots. Postcode 4570 (Southern Region) records the highest number of fatalities, followed by Bundaberg (4670), and several Central Queensland postcodes like 4702 and 4704. This highlights the need for interventions not only in urban centres but also across regional and rural highways where severe crashes remain prevalent.

Table 1 complements these spatial visualisations by summarizing fatality counts across Queensland's Transport Regions. The Southern Region contains multiple high-fatality postcodes, while SEQ South and Central Queensland also contribute significantly. Presenting the data by region helps policymakers allocate safety resources more effectively, prioritizing regions where fatal crashes are most concentrated.

Finally, Figure 6 shows a time series for the top 10 high-risk postcodes from 2001 to 2024. While fluctuations are observed year to year, a gradual long-term decline in fatalities is evident. Peaks between 2005 and 2008 exceeded 55 fatalities annually, followed by notable reductions. However, periodic rebounds in recent years (e.g., 2019, 2023) indicate that these hotspot locations continue to require ongoing monitoring and targeted interventions.

Overall, while long-term improvements are evident, persistent regional hotspots emphasize the need for location-specific safety measures addressing rural road design, speed enforcement, and emergency response capabilities.

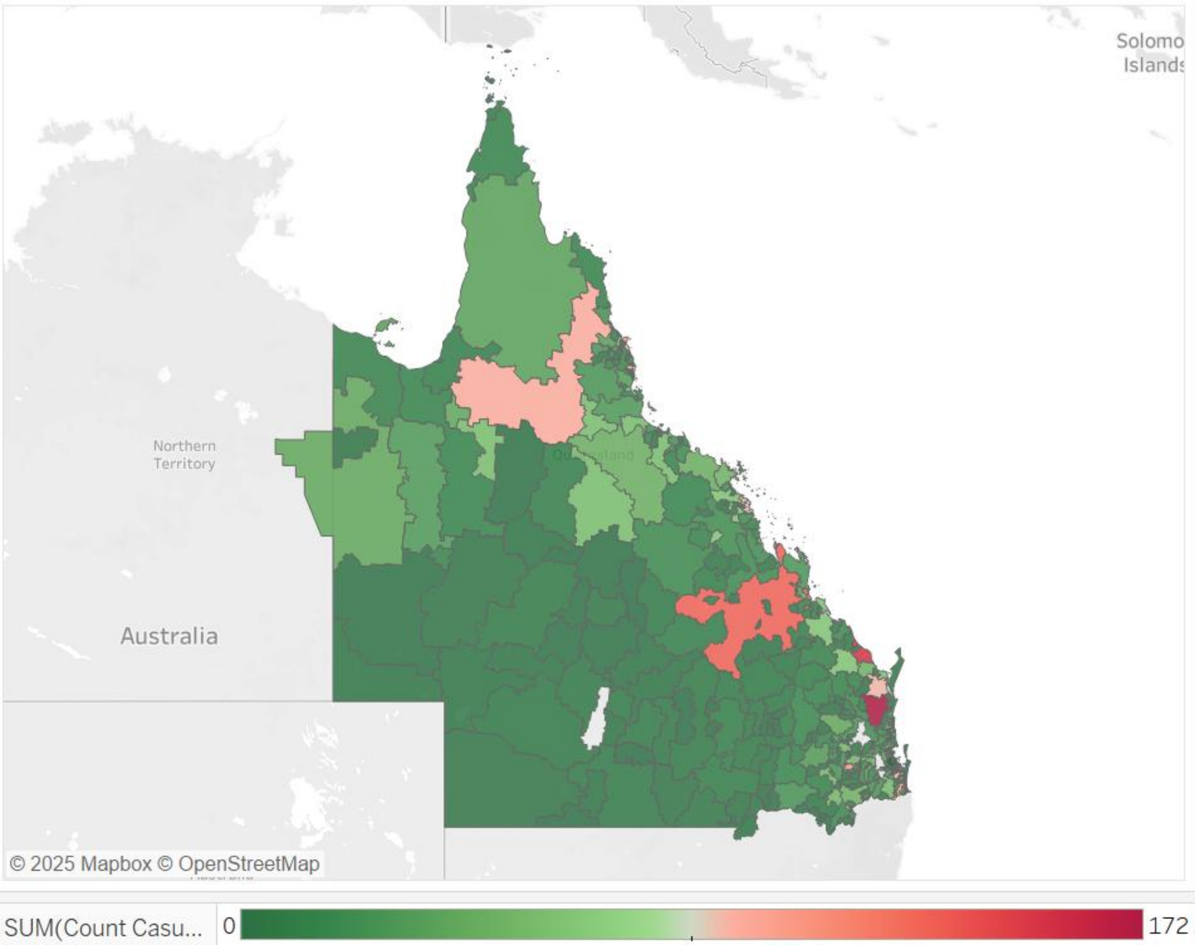

Figure 4: Fatal Crash Casualties Count by Queensland Post Code

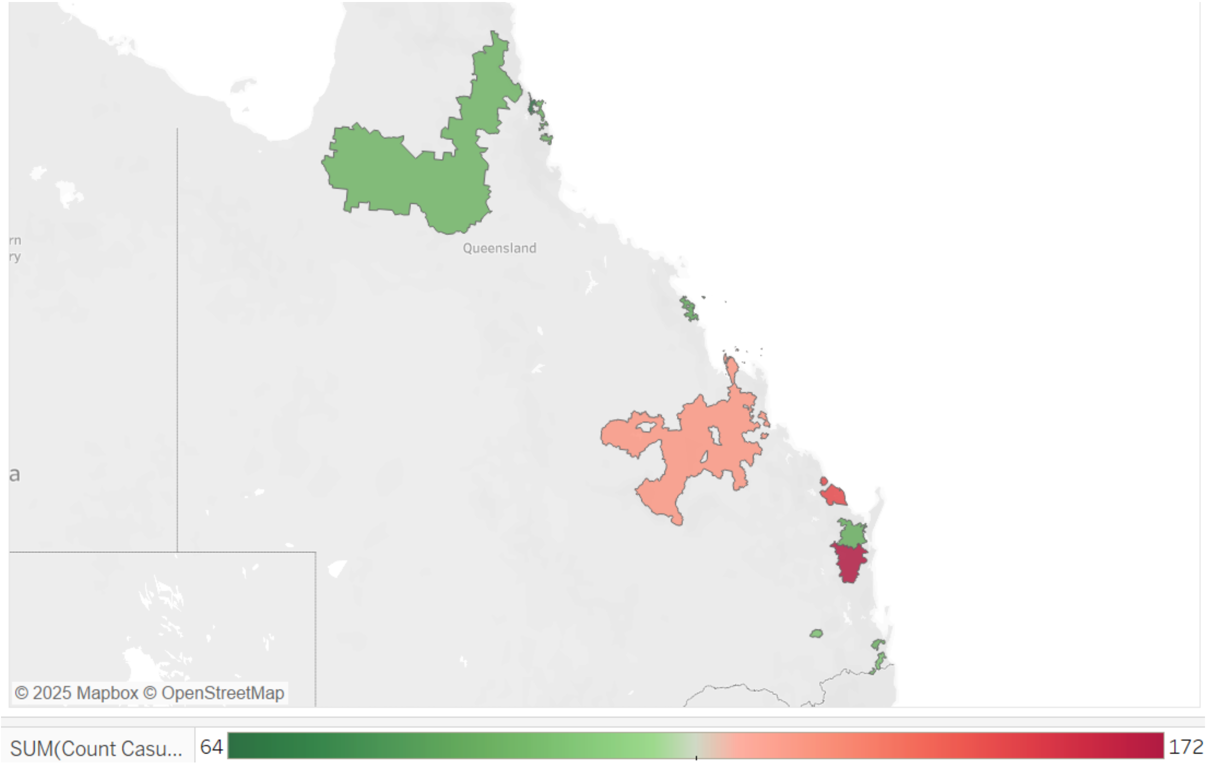

Figure 5: Top 10 Postcodes by Fatal Crash Casualties

Table 1: Fatal Casualties Count per Post Code and Transport Region

Loc Queensland Transport Region	Loc Post Code	
Central	4740	90
	4702	131
Northern	4870	64
	4871	95
SEQ South	4211	102
	4207	94
Southern	4670	153
	4350	101
	4570	172
	4650	92

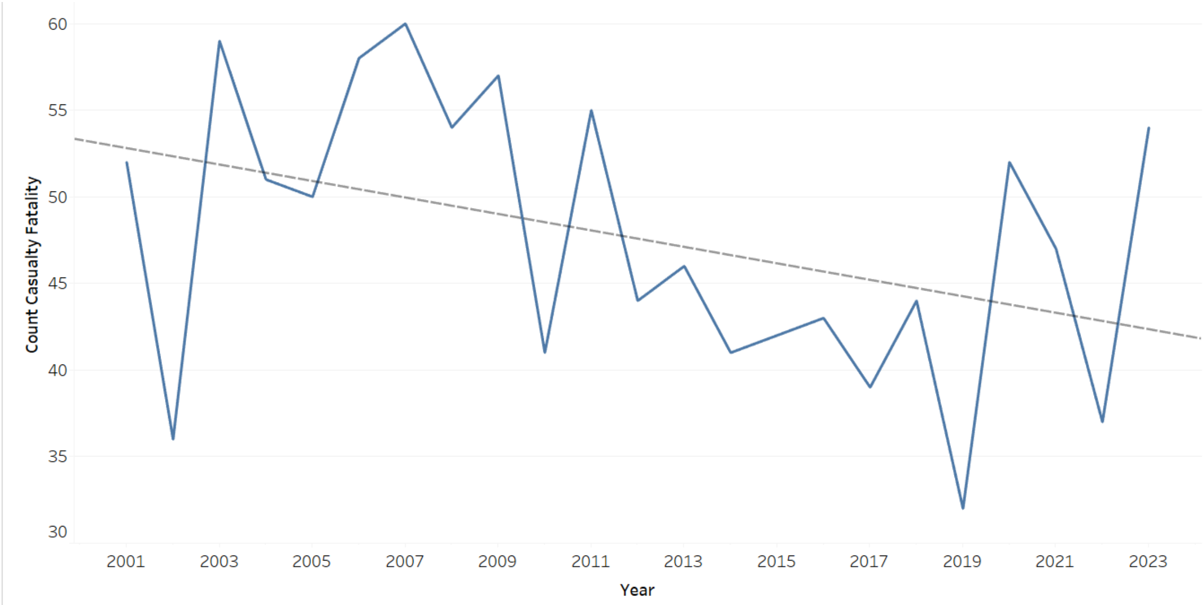

Figure 6: Fatal Crash Casualties Time Series -- Top 10 High-Risk Postcodes

3.3. Crash Severity Patterns Across Time and Contributing Factors

To analyse how crash severity varies across time periods and contributing factors, three visualisations were developed that explore hourly patterns, crash types, and severity distribution.

Figure 7 presents a stacked bar chart showing the total number of crashes at each hour of the day, segmented by severity levels. This visualisation was selected to capture both overall crash volume and how severity changes throughout the day. Crash volume rises sharply after 5 AM, peaking between 3 PM and 5 PM, corresponding with daily work commutes and school pickups. These high-volume periods are dominated by minor injuries and medical treatment cases, consistent with congestion-related crashes that typically result in less severe outcomes. Late night and early morning hours (0–5 AM) show lower crash volumes but still contribute to serious outcomes, although their relative share remains stable across the day.

Figure 8 examines severity distribution across the top 10 crash types, using disaggregated casualty counts rather than overall crash severity labels. Head-on collisions display the highest proportion of fatalities, emphasizing their severe consequences. Off-carriageway crashes and pedestrian incidents also show elevated fatality shares. In contrast, rear-end collisions account for the majority of minor injuries and medical treatments due to their frequency but generally lower severity. Intersection and lane change crashes also predominantly result in minor injuries, while opposing vehicle turning crashes contribute significantly to hospitalisations and medical treatments. These distinct severity profiles highlight the need for crash-type-specific interventions, such as centreline barriers, pedestrian protections, and collision avoidance technologies.

Figure 9 presents a proportional heatmap showing how the distribution of each severity level varies across the hours of the day. Minor injuries and medically treated crashes are more concentrated in afternoon peak hours (14:00–16:00), while hospitalisations and fatalities are slightly more evenly spread throughout the day. These patterns suggest that severe crashes may not be solely tied to traffic volume but influenced by additional behavioural and environmental factors.

Overall, severity patterns in Queensland vary across both time and crash type. While most daytime crashes lead to minor injuries, certain crash types and contributing factors continue to drive severe outcomes, supporting the need for targeted and data-driven safety interventions.

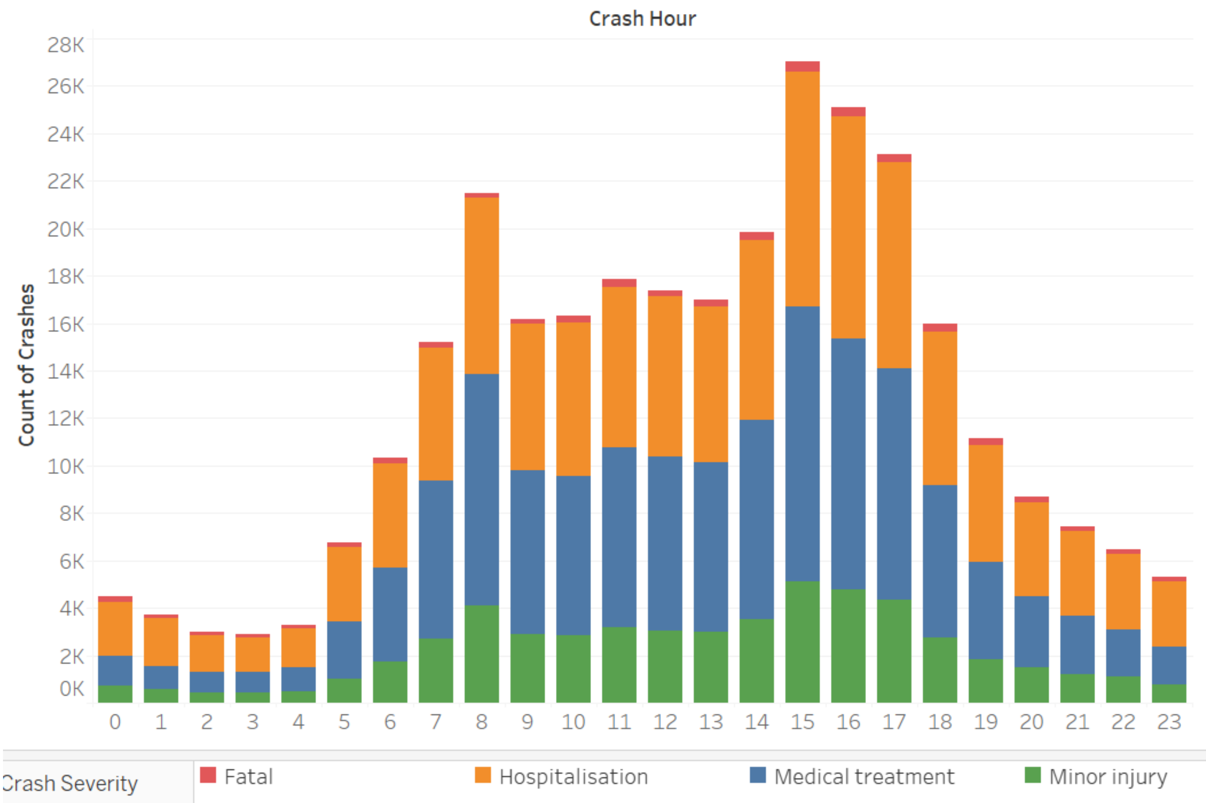

Figure 7: Crashes by Severity by Hour

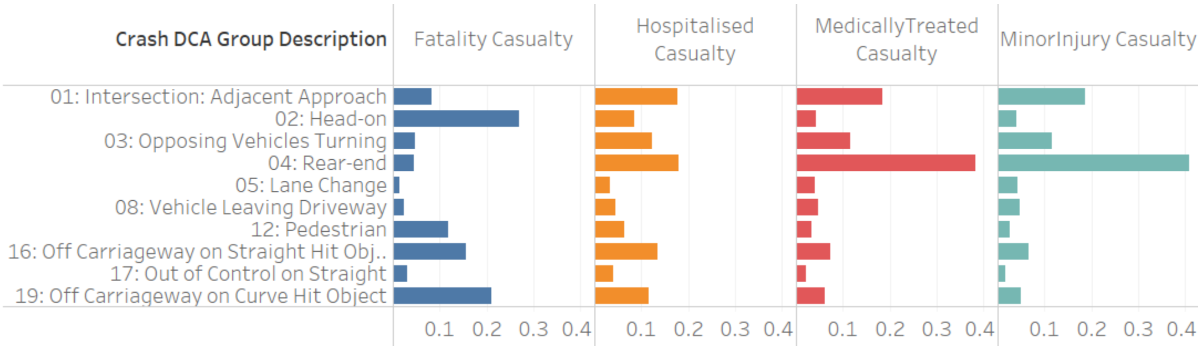

Figure 8: Crash Type Severity Distribution Across Top 10 Crash Causes

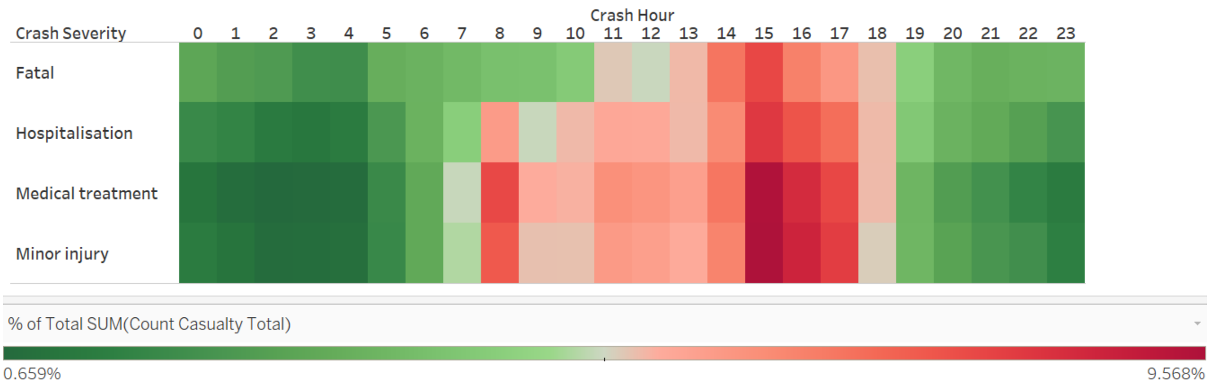

Figure 9: Hourly Crash Severity Proportions Normalized by Crash Volume

4. Ethical Principles in Data Visualisation

Throughout this project, key ethical principles were applied to ensure responsible, transparent, and meaningful visualisations.

4.1. Accuracy and Data Integrity:

The dataset was carefully cleaned to ensure accuracy. Incomplete records, such as unknown locations and inconsistently recorded property damage, were removed to prevent misleading insights. Maintaining accurate data was essential to avoid flawed conclusions that could negatively affect public safety decisions.

4. 2. Fairness and Avoidance of Bias:

Consistent processing was applied across crash types, severity levels, and time periods.

Extreme cases that might exaggerate trends were avoided, ensuring a balanced representation that included both high-frequency minor crashes and more severe but less frequent incidents.

4.3. Common Good and Transparency:

The visualisations were designed to support policymakers by providing clear insights into crash severity patterns across time, location, and crash types. These insights aim to guide targeted interventions, promoting public safety and contributing to the community's well-being. Applying these principles ensured trustworthy, ethical analysis.

5. Conclusion

The analysis of Queensland's road crash data reveals several key patterns. While total crashes have declined, fatalities remain stable and concentrated in specific high-risk postcodes. Afternoon commutes account for most minor injuries, while late-night and early morning crashes show higher severity due to fatigue, alcohol, or speed. Head-on, off-carriageway, and pedestrian crashes lead to more severe outcomes, whereas rear-end and intersection crashes mainly cause minor injuries. These insights highlight the need for behavioural interventions and engineering solutions targeting high-risk times and crash types. Ethical principles ensured the analysis remained fair, accurate, and transparent to support effective safety decisions.